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Aditya ML, NLP, DL
Course

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Sec - 25

Introduction to ML

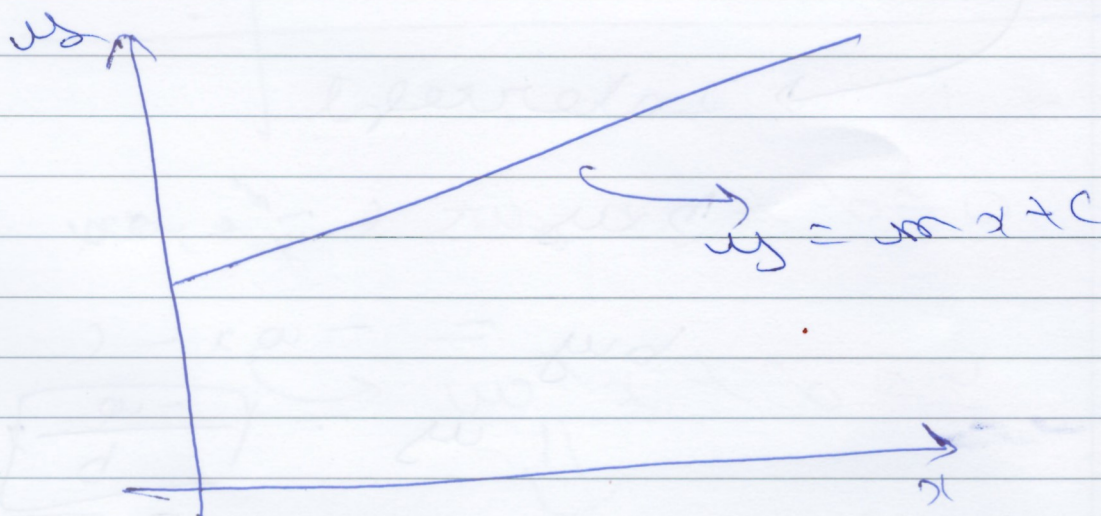
lec-3 Equation of
Line, 3D & Hyperplane

lec-4 Distance of a
point from plane

lec-5 Instance based
vs Model Based
Learning

Equation of Line, 3-D plane
→ Hyperplane

(n-dimensions)



some other notations
as:-

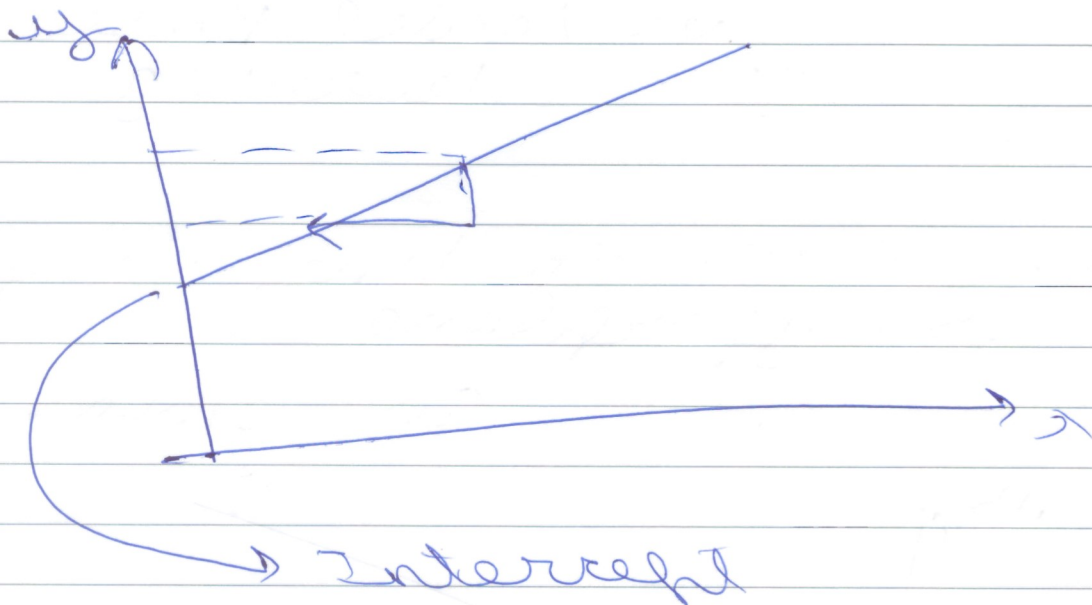
$$y = \beta_0 + \beta_1 x$$

$$ax + by + c = 0$$

→ Both equations are
almost same.

→ $m = \text{slope}$

$c = \text{Intercept}$



$$ax + by + c = 0$$

$$by = -ax - c$$

$$y = \frac{-ax - c}{b}$$

(3)

same equation obtained
as above

$$y = \boxed{\frac{-a}{b}} x - \boxed{\frac{c}{b}}$$

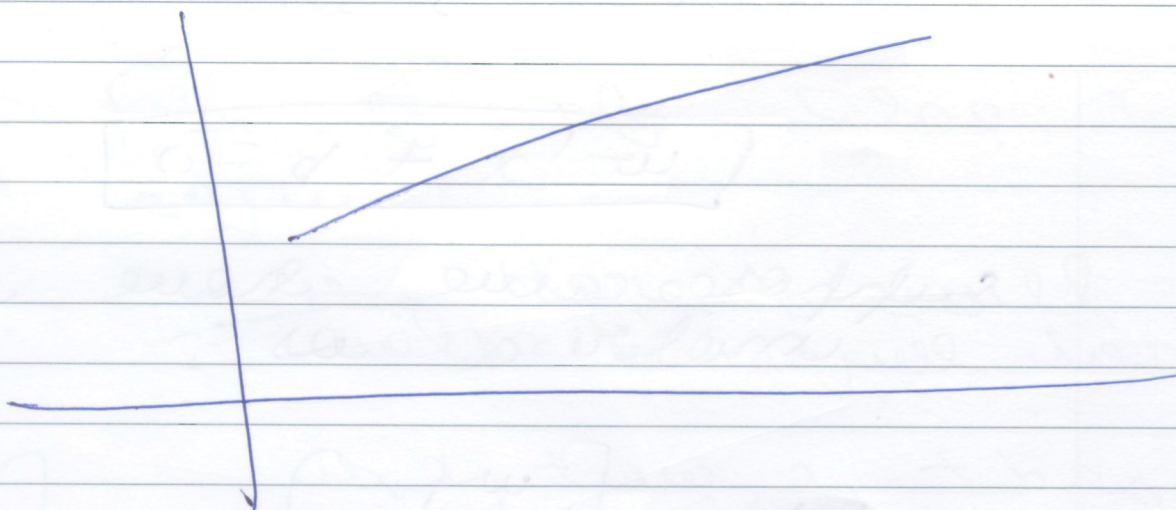
$$y = mx + c$$

m

c

$$y = mx + c$$

now let us see and
more equations



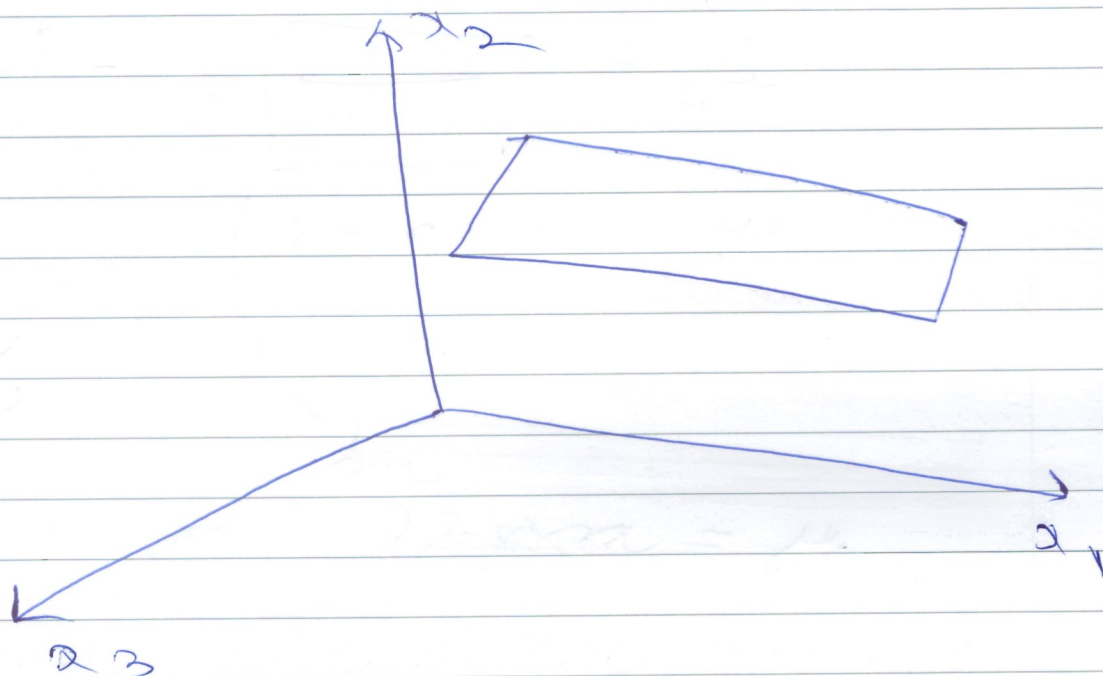
$$w_1 x_1 + w_2 x_2 + b = 0$$

$$\hookrightarrow w^T x + b = 0$$

Equation of a
straight line

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Now suppose we have 3 axes as:-



$$w_1 x_1 + w_2 x_2 + w_3 x_3 + b = 0$$

$$w^T x + b = 0$$

Suppose we have a vector as:-

$$w = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} \quad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

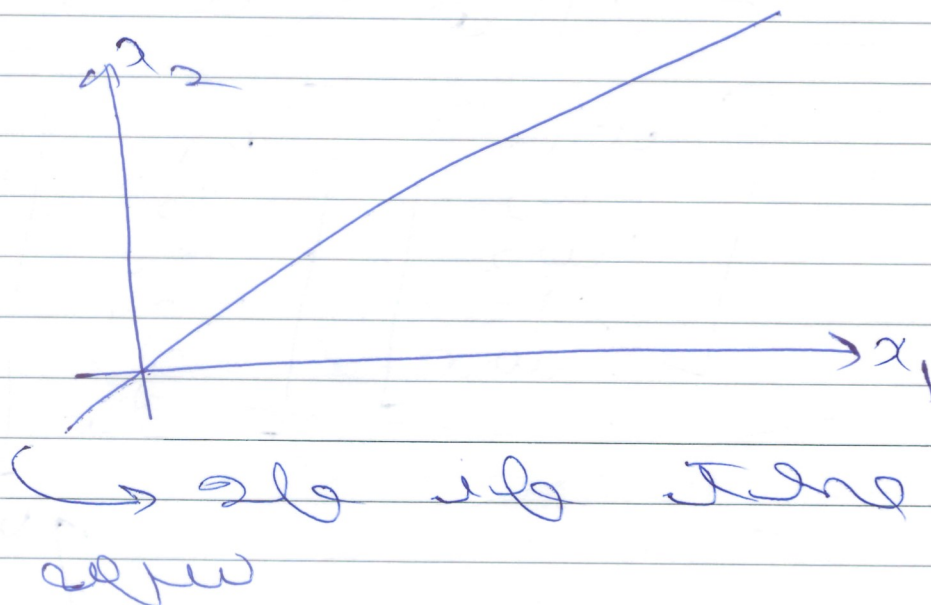
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n - Dimension plane

$$w_1 x_1 + w_2 x_2 + w_3 x_3 + \dots + w_n x_n = 0$$

$$w^T x + b = 0$$

let's consider one case



~~the~~ Here b will also be equal to 0

$$w_1 x_1 + w_2 x_2 + b = 0$$

↓

$$w_1 x_1 + w_2 x_2 = 0$$

$$w^T x = 0$$

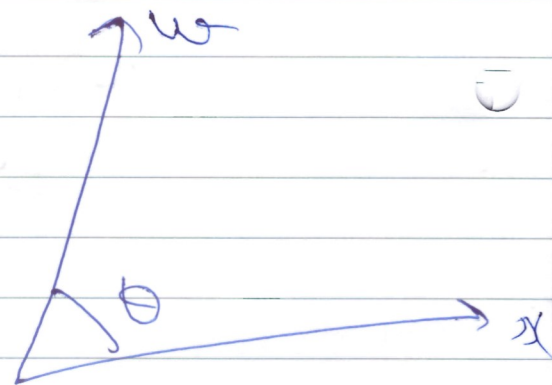
Equation of a straight line passing through origin :-

$$w^T x = 0$$

Equation of a plane

$$\Rightarrow \uparrow x : w^T x = 0$$

$$\begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ \vdots \\ w_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ \vdots \\ x_n \end{bmatrix}$$



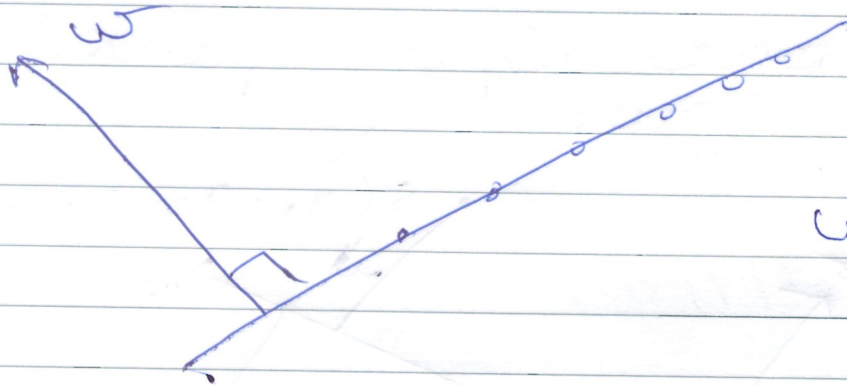
$$w \cdot x = w^T x = \|w\| \|x\| \cos \theta$$

= 0

$$\theta = 90^\circ$$

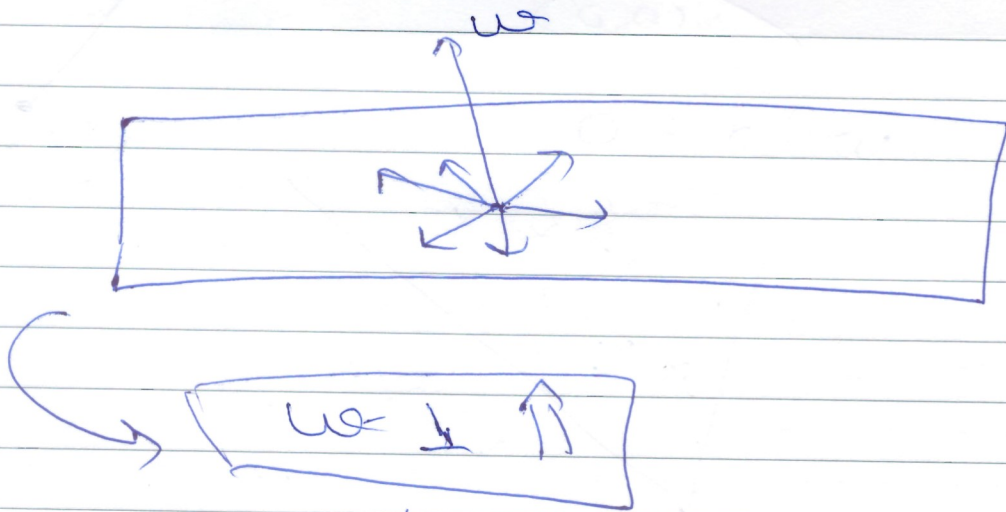
$$\cos \theta = 0 \quad \text{Ans}$$

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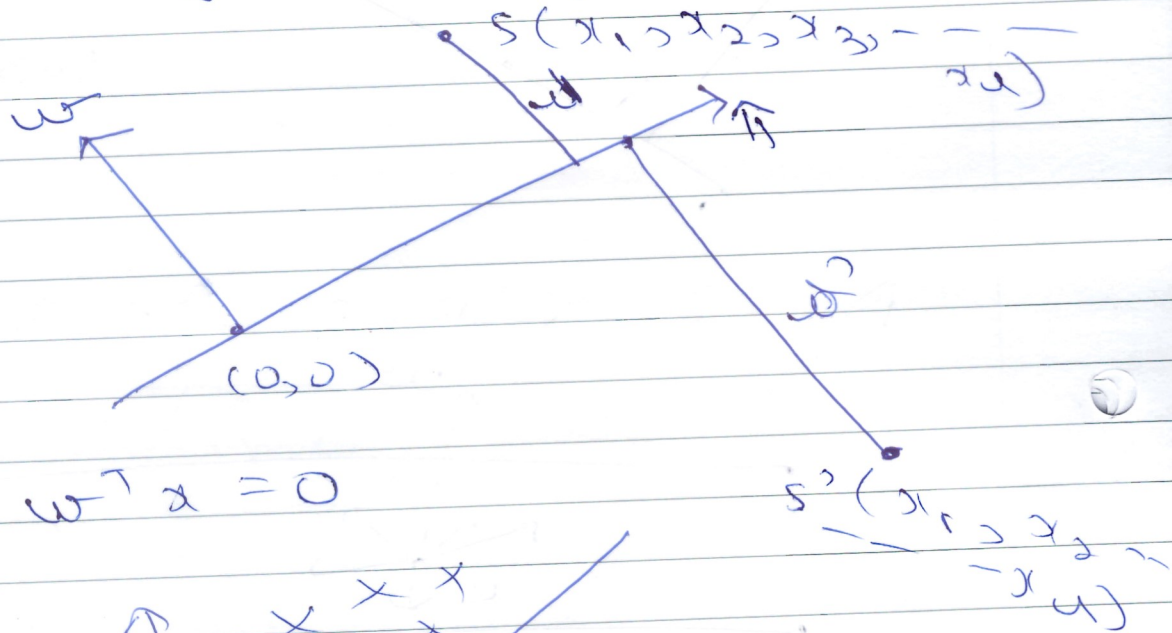
$$\theta = 90^\circ$$
$$\cos \theta = 0$$

A 3D plane looks like



but in this
equation the
represent the sign.

=> Distance of a point from plane



$wb = w^T s \rightarrow s$ is basically that point

one equation of the distance is :-

$$d = \frac{w^T s}{\|w\|}$$

(9)

$$\Rightarrow w^T s = \|w\| \|s\| \cos \theta$$

$$w^T s = \|w\| \|s\| \cos \theta$$

$$w^T s = \|w\| \|s\| \cos \theta$$

$$= +ve$$

Now alg tells about θ

$$w' = \frac{w^T s}{\|w\|}$$

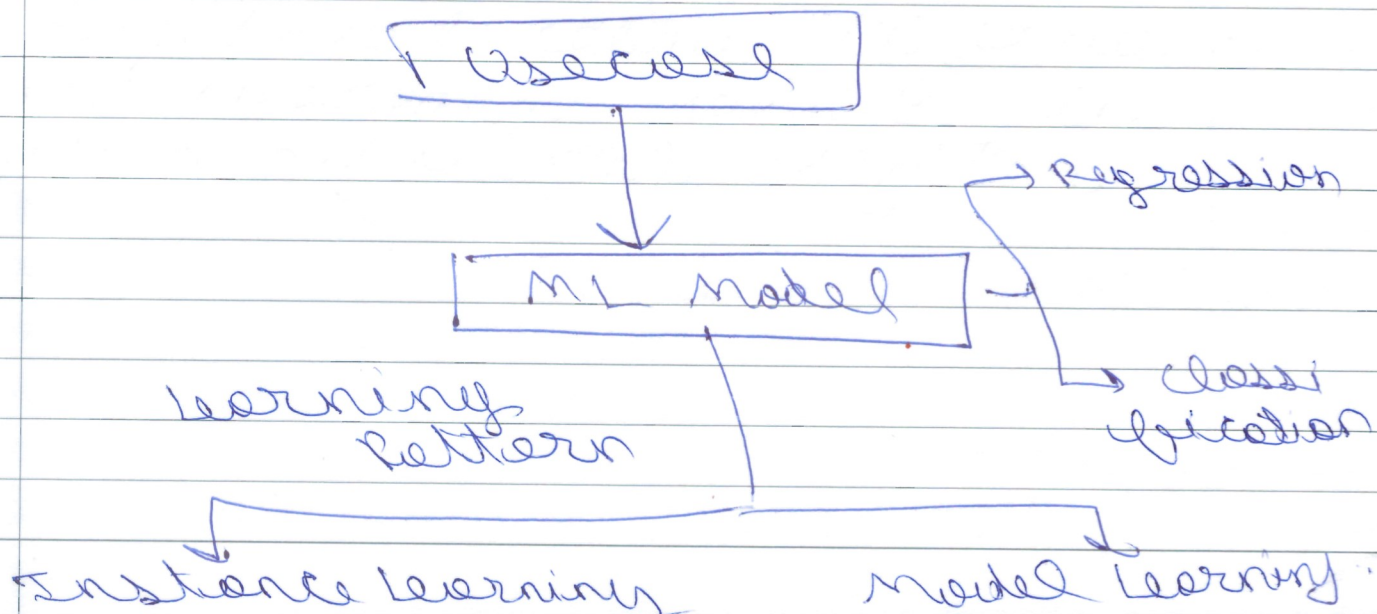
$$w^T s' = \|w\| \|s'\| \cos \theta$$

$$\Rightarrow -ve$$

Instance Based Learning

vs

Model Based Learning



- The entire difference between supervised and unsupervised learning is based on learning about pattern of data.

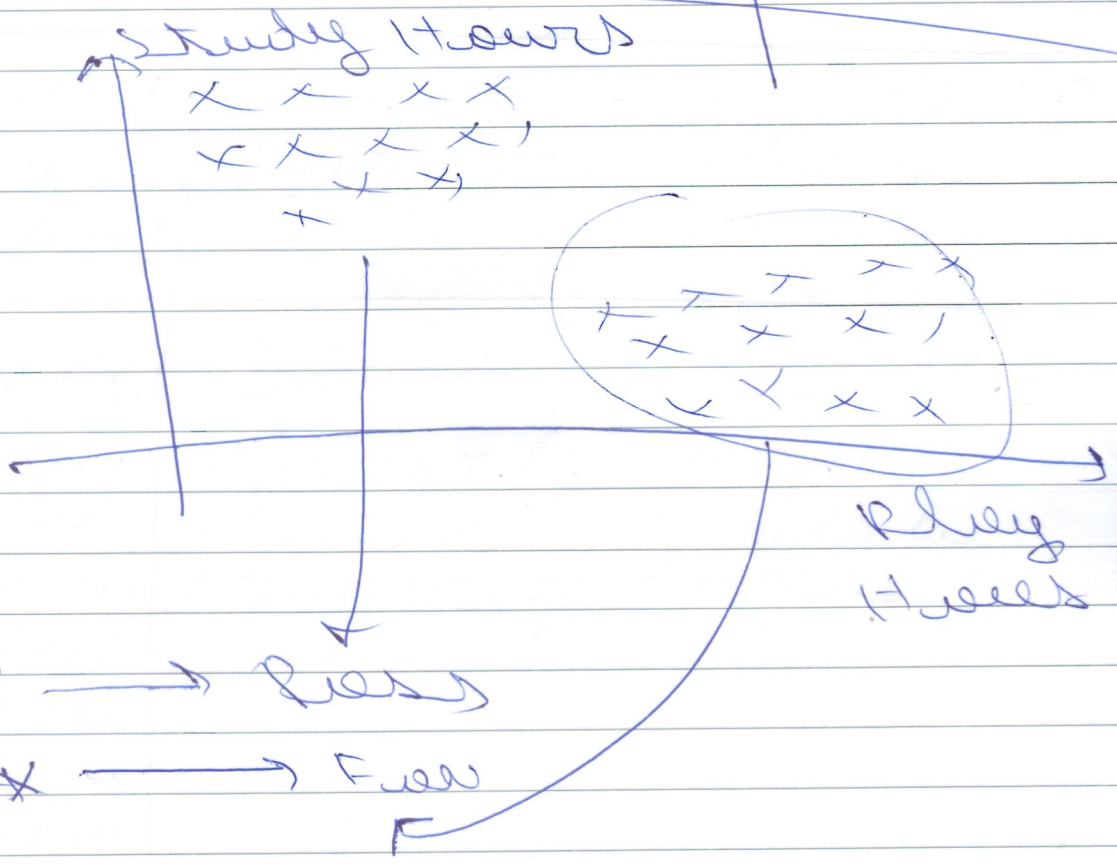
- If the model is completely dependent on input,

→ This happens in Instance Based learning
→ The output is based on the data.

- In Model Based learning the ~~data~~ model tries to understand the pattern within the data.

- Once it understands pattern, it creates a generalization, which is used for the analysis.

No-obj play	No-obj study	Loss/ Fuel
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Instance Based Learning



Domain Expert



It mainly uses KNN

Model

Instance based is better if more data is memorized than the data.

Usual machine learning

Instance based learning

(1) prepare for model training.

(1) prepare the data for model training.

(2) train model using training data to estimate model parameters.

(2) do not train. model. often discarded. hashed. until the training query is received.

(3) stores data in suitable form.

(3) there is no model to store.

